Guillermo Fernando Farfán Romero

Senior Software Engineer & Cloud Solutions Architect

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PROFESSIONAL SUMMARY

Senior Software Engineer, Cloud Solutions Architect, and Technical Lead with 8+ years of experience designing, developing, observing, and scaling distributed enterprise systems and mission-critical cloud infrastructures. Specialist in **Google Cloud Platform (GCP)**, backend engineering with **Python 3.10+ (FastAPI, Asyncio, Celery, SQLAlchemy 2.0, Pydantic v2)**, **Java (Spring Boot)**, and **TypeScript / Node.js**, orchestration on **Kubernetes (GKE Autopilot)**, Infrastructure as Code (**Terraform**), and advanced data persistence (**PostgreSQL with pgvector, SQL Server, BigQuery, and Redis**).

Proven track record leading event-driven architectures (EDA), multi-cloud automation with Terraform, GenAI/LLM integrations, distributed observability with **OpenTelemetry, Prometheus, and Grafana**, and cloud financial optimization (**FinOps**).

TECHNICAL SKILLS & COMPETENCIES

- **Backend Development & APIs:** Python 3.10+ (FastAPI, SQLAlchemy 2.0, Pydantic v2, Asyncio, Celery, Django REST Framework), Java (Spring Boot, Microservices), TypeScript, Node.js, RESTful APIs, Streaming APIs (SSE / WebSockets), GenAI / LLM Integration (Vertex AI, Gemini / OpenAI API, RAG, pgvector), Clean & Hexagonal Architecture, DDD.
- **Cloud, Virtualization & DevOps:** Google Cloud Platform (GKE Autopilot/Standard, Compute Engine, Cloud Run, Migrate to VMs - M4CE, Cloud Storage, Shared VPC, Zero-Trust IAM, Cloud Armor, Cloud KMS), Microsoft Azure, Microsoft Hyper-V, VMware vSphere, Nutanix AHV, Veeam Backup & Replication (VBR), Terraform, Terragrunt, Docker, ArgoCD, GitHub Actions, GitLab CI.
- **Databases & Data Platforms:** PostgreSQL (Declarative Partitioning, pgvector, PgBouncer), Microsoft SQL Server (2012–2022, AlwaysOn, Performance Tuning), SAP HANA IaaS Infrastructure, Redis (Distributed Caching, Pub/Sub), Google BigQuery, MySQL, MongoDB, Alembic, Apache Airflow.
- **Observability, Quality & SRE:** OpenTelemetry (OTel), IBM Instana APM, Zabbix, Google Cloud Operations, Prometheus, Grafana, TDD (Pytest, Jest), SonarQube, Trivy, DORA Metrics, SLOs/SLIs, FinOps, Architecture Decision Records (ADRs).
- **Systems, Storage & Networking:** Linux (SLES for SAP, RHEL, Ubuntu, Debian), Windows Server (Active Directory), Fibre Channel SAN Storage (MPIO / CSV), Switch Embedded Teaming (SET), Infoblox (DNS/DHCP), Load Balancing, and DRP/BCP.

PROFESSIONAL EXPERIENCE

Senior Python Backend & Financial Automation Engineer | Reforest Latam

May 2026 – Present | Full-time (Remote)

- Architected and developed end-to-end the financial operations platform and transactional Telegram bot in **Python 3.11+ (aiogram 3.x, asyncio, Pydantic)** with **Redis-persisted FSM** state management (RedisStorage and events_isolation double-click protection), shared HTTP connection pooling (aiohttp), exact decimal accounting (ROUND_HALF_UP), real-time **fail-closed RBAC**, and **Baserow / Google Drive API** integration.
- Built an **atomic multi-currency bank reconciliation engine** featuring native PDF/CSV/Excel statement parsers (BBVA, Galicia, Macro, Mercury, Balanz), **SHA-256** deduplication in Redis, decoupled staging with semantic auto-classification, batch injection with **atomic batch rollback**, historical **MEP Dollar** exchange rate lookup with T-1 to T-5 business-day fallback, and sanitized **ARCA** tax exports.
- Implemented **dual LLM OCR & automated classification (**Gemini 2.5 Flash **and** DeepSeek **via OpenRouter)** using semaphore-bounded concurrent PDF chunking and prompt-injection sanitization; deployed on **Linux VPS with Docker Compose**, automated daily Drive backups, and **GitHub Actions CI/CD** running **Ruff** and **514 passing Pytest integration tests (100% green)**.

Senior Cloud Architect & Software Consultant | Independent Consulting & Enterprise Projects

January 2021 – Present | Strategic B2B Technology Consultant (Independent / Contract / Remote)

- **Multi-Cloud Architecture, Virtualization & IaC Automation (Enterprise Clients under NDA):** Designed and provisioned multi-environment infrastructures on **Google Cloud Platform and Microsoft Azure** using reusable **Terraform** modules (Landing Zones, Shared VPC, GKE, Cloud KMS, M4CE), corporate server migrations and high-availability IaaS infrastructure on **Linux (SLES, RHEL) and Windows Server**, and **Microsoft Hyper-V / VMware** virtualization clusters with **Fibre Channel SAN storage and Veeam VBR**.
- **Backend Engineering & Fintech Platforms (Clients under NDA):** Built high-concurrency **FastAPI** middleware, asynchronous pipelines with **SQLAlchemy 2.0, Pydantic v2, Celery, and Redis (>60% p95 latency reduction)**, low-latency REST APIs in **Django REST Framework**, credit-scoring microservices, and telemetry services in **Java (Spring Boot)** on GCP.
- **Distributed Observability & SRE Resilience (Clients under NDA):** Implemented end-to-end distributed tracing and APM monitoring with **OpenTelemetry, IBM Instana APM, Zabbix, Prometheus, and Grafana**, achieving a **45% reduction in MTTR**, while designing Disaster Recovery (DRP/BCP) and High Availability strategies for mission-critical databases.

Senior Software Engineer & Database Administrator (DBA) | University of Buenos Aires (UBA)

March 2018 – February 2024 | Institutional Systems Specialist (Part-time / Buenos Aires, Argentina)

- Spearheaded performance tuning (execution plans, re-indexing, deadlock resolution) and High Availability across **8+ TB of institutional data (+500 GB migrated from critical legacy databases)** on **SQL Server and PostgreSQL** serving **+50,000 users (+10,000 concurrent active users)**, cutting **p95 query latency by 55% (-40% average response time)** under **RPO < 15 min and RTO < 1 hour**.
- Built asynchronous microservices, **Python/Alembic ETL pipelines (-60% manual operational effort)**, and **React/TypeScript** web interfaces for real-time reporting.

Systems Infrastructure & Operations Coordinator | Government of the City of Buenos Aires (GCBA)

January 2020 – December 2023 | Ministry of Economic Development (Full-time / Buenos Aires, Argentina)

- Coordinated hybrid infrastructure, maintenance windows, and administration of **SQL Server, PostgreSQL, MySQL, and MongoDB** across citizen portals serving **100,000+ monthly users**, achieving **99.9% SLA**, **-30% production incidents**, **-35% backup windows**, **-40% backend data inconsistencies**, and training **+30 officials**.

FEATURED PROJECTS & HACKATHONS

- **Autonomous FinOps Engine (Aleph Hackathon 2026):** Autonomous Python engine that detects idle GCP infrastructure and automatically generates **Terraform (IaC)** remediation code.
- **OmniSync (EDA) & Monitor CI/CD (DORA Metrics):** Polyglot event-driven microservices engine with eventual consistency and Dead Letter Queues (DLQ), and a real-time TypeScript/Node.js/WebSockets dashboard integrated with GitHub Actions and GitLab CI.

EDUCATION, CERTIFICATIONS & LANGUAGES

- **Education: Associate Degree in Information Systems** | UTN *(In Progress — Expected Graduation: Dec 2027)* · **Bachelor's Studies in Labor Relations & HR** | UBA *(Advanced University Coursework Completed / Organizational Leadership)*.

  - *Certifications* (Verifiable on Credly & Coursera):* IBM DevOps and Software Engineering Professional Certificate | IBM Full Stack Software Developer (V5) | IBM Java Developer (V2) | IBM JavaScript Backend Developer (V2) | Meta Full-Stack Engineer & Meta Back-End Developer | Google Data Analytics | IBM IT Scrum Master | Google Cloud Fundamentals.
- **Languages: Spanish:** Native · **English:** Professional Working Proficiency (B2 — Fluent technical communication for live engineering syncs, RFCs, and documentation).